

KEMCHUG, Russian Federation

WMO: 295620

Lat: 56.0936N

Lon: 91.7166E

Elev: 1089

StdP: 14.13

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-26.7	-38.2	0.6	-32.1	-32.3	0.9	-26.4	19.2	13.0	16.8	13.7	1.5	30	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	83.0	65.4	79.6	63.9	76.6	62.7	68.0	77.3	66.2	75.8	64.5	73.7	5.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	95.4	71.9	62.8	88.9	70.3	61.0	83.5	68.8	33.1	77.4	31.6	76.0	30.3	73.9	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.8	14.6	12.9		-36.5	88.5	8.5	2.4	-42.6	90.3	-47.6	91.7	-52.4	93.0	-58.6	94.8	
				-36.6	71.3	8.5	2.5	-42.7	73.1	-47.7	74.5	-52.4	75.9	-58.6	77.7	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.1	2.3	7.1	22.5	35.6	48.2	61.4	63.9	59.2	47.9	34.6	18.0	6.5	(6)
(7)		DBStd	24.17	15.84	14.05	11.32	9.91	10.15	7.45	5.50	6.65	8.14	9.96	13.48	14.85	(7)
(8)		HDD50	7079	1478	1200	853	442	161	9	0	8	135	486	959	1348	(8)
(9)		HDD65	11411	1943	1620	1318	883	528	150	85	204	514	944	1409	1813	(9)
(10)		CDD50	1269	0	0	0	9	106	352	431	292	72	7	0	0	(10)
(11)		CDD65	125	0	0	0	0	7	43	51	23	1	0	0	0	(11)
(12)		CDH74	1250	0	0	0	2	84	485	432	222	25	0	0	0	(12)
(13)		CDH80	251	0	0	0	0	16	109	79	46	1	0	0	0	(13)
(14)	Wind	WSAvg	5.6	5.3	5.7	6.7	6.8	6.6	4.8	4.1	4.3	5.2	6.0	6.3	5.8	(14)
(15)	Precipitation	PrecAvg	21.1	0.9	0.7	0.8	1.2	1.8	2.8	2.6	3.0	2.4	1.9	1.7	1.3	(15)
(16)		PrecMax	25.9	1.6	1.5	1.6	2.2	3.9	5.4	5.4	6.3	4.4	3.1	3.1	2.3	(16)
(17)		PrecMin	15.8	0.1	0.2	0.1	0.4	0.5	1.0	0.7	1.0	0.6	0.7	0.8	0.4	(17)
(18)		PrecStd	2.5	0.4	0.3	0.4	0.5	0.9	1.1	1.2	1.2	1.0	0.6	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.3	37.3	51.8	66.5	81.8	87.5	86.2	85.1	77.6	63.6	48.8	34.6	(19)
(20)			MCWB	29.1	30.5	39.1	49.2	60.3	64.7	68.1	68.3	61.0	50.0	41.9	29.2	(20)
(21)		2%	DB	28.7	32.6	45.6	60.7	75.4	83.1	82.3	79.7	70.9	57.9	42.7	30.8	(21)
(22)			MCWB	25.4	27.8	36.1	46.7	56.5	64.1	66.4	65.5	57.8	47.4	37.6	27.3	(22)
(23)		5%	DB	25.8	29.8	41.3	55.9	70.8	79.6	78.7	75.6	65.9	52.9	38.6	28.1	(23)
(24)			MCWB	22.7	25.8	33.8	44.0	54.2	62.9	64.9	63.6	55.2	44.9	33.7	25.4	(24)
(25)		10%	DB	22.9	26.5	37.7	51.5	66.0	76.0	75.8	71.7	61.4	48.9	34.5	25.1	(25)
(26)			MCWB	20.1	23.3	31.5	41.5	51.9	61.4	63.9	61.1	52.8	43.0	31.1	22.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.9	31.8	40.4	50.6	61.6	68.2	72.0	70.4	62.0	52.0	42.5	30.3	(27)
(28)			MCD B	32.9	35.7	49.7	62.8	80.1	78.7	81.6	81.6	71.6	60.9	47.6	32.6	(28)
(29)		2%	WB	26.0	28.6	36.9	48.1	58.2	66.5	68.8	66.8	58.6	48.6	38.1	27.8	(29)
(30)			MCD B	28.4	31.9	44.1	59.3	71.2	78.1	78.0	76.6	68.5	56.0	42.7	30.5	(30)
(31)		5%	WB	22.8	25.9	34.3	44.9	55.8	64.8	66.8	64.5	56.3	46.1	34.0	25.6	(31)
(32)			MCD B	25.5	29.4	40.5	54.6	67.5	76.2	75.3	73.0	64.0	52.6	38.1	28.1	(32)
(33)		10%	WB	20.0	23.3	31.8	41.7	53.2	63.1	65.2	62.5	54.1	42.9	31.4	22.5	(33)
(34)			MCD B	22.8	26.7	37.3	50.7	64.4	73.5	73.3	69.8	60.1	48.2	34.1	25.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	15.4	15.9	16.2	18.2	19.3	17.2	17.1	15.4	11.3	11.1	12.1	(35)
(36)			MCDBR	10.1	13.3	19.0	23.4	25.6	26.0	23.0	23.9	24.0	18.4	11.5	10.1	(36)
(37)			MCWBR	8.9	10.8	12.4	13.6	12.8	11.1	10.2	11.4	12.6	11.0	7.8	8.8	(37)
(38)		5% WB	MCD BR	9.5	12.3	16.9	22.0	22.1	21.2	18.9	20.7	20.8	17.5	10.7	10.2	(38)
(39)			MCWBR	8.8	10.5	11.6	13.7	12.1	10.5	9.8	10.7	11.7	11.0	7.8	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.284	0.317	0.365	0.396	0.411	0.406	0.428	0.410	0.371	0.351	0.317	0.285	(40)	
(41)		taud	2.005	1.985	1.927	2.070	2.194	2.301	2.252	2.273	2.329	2.222	2.032	1.952	(41)	
(42)		Ebn at Noon	182	224	243	258	262	266	256	252	246	214	164	143	(42)	
(43)		Edh at Noon	21	32	44	45	42	38	40	36	30	26	21	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	248	557	984	1367	1568	1879	1682	1343	870	449	222	148	(44)	
(45)		RadStd	29	46	82	93	172	162	105	119	110	40	31	20	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page